

Using the 21-cm Interstellar Hydrogen Line to study the galaxy UGC 328

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ABSTRACT

Hydrogen is by far the most abundant and important element in our universe. Since the discovery of radio astronomy in the mid 20th century, humanity has continued to gather knowledge about the importance of neutral hydrogen (HI). The discovery of the 21-cm hydrogen line helped to bolster research revolving around Galaxies and their respective structures. Galaxy UGC 328 is one of the multitude of Galaxies in which the 21-cm neutral hydrogen line has assisted in the study of it. This study is located in the reference paper (Pisano et al. 2002). I present a comparison study of the Galaxy UGC 328 in which I will use spectral line analysis techniques to achieve my own results to compare with the results shown in (Pisano et al. 2002).

1 INTRODUCTION

In the mid 20th century, astronomers were confident that the space in between stars was not a perfect vacuum. Using earth-bound spectrometers, astronomers were able to show the existence of interstellar ionized hydrogen near stars (Wakelam et al. 2017). However, non-ionized hydrogen is invisible at the optical and near-infrared wavelengths, thus, astronomers were left with an incomplete grasp on hydrogen in the universe. This was the case until Jansky and Reber's pioneering work with radio astronomy allowed astronomers to use this new technology to detect the 21-cm emission spectrum (Carilli 2006) of neutral hydrogen located in the Galactic gravitational potential. In this paper, I will show the HI studies of UGC 328. I will show how I completed the study along with the results with their respective uncertainties. The layout of the rest of this paper is as follows: in §2 I discuss how the study was done and how someone else could recreate it. The results of the study are given in §3. In §4, I discuss the results of the study. Finally, in §5, I present the conclusions.

2 METHODS

To first start the HI studies of UGC 328, I first extract the radio astronomical data in the form of radio frequency verses intensity. In this study, the file UGC328.dat was used from the reference paper (Pisano et al. 2002). In the data, there are two different columns for intensity, one taken on source and one taken off source. First, I made a plot of the frequency versus the on-source and off-source intensity. I used this plot to compare and get a feel for what these values represent. Using these values, I wrote code that subtracts the off-source intensity from the on-source intensity. This subtraction helps to separate the source intensity from the other data. I call this calculated value sub-source intensity. After getting the sub-source intensity, I took the Doppler formula to transform the frequency data into a velocity scale

$$v = -c \frac{f - f_0}{f_0} \quad (1)$$

where v is the velocity values, c is the speed of light, f is the actual frequency, and f_0 is the observed frequency. Once I got this plot, I

Table 1. A comparison of the resulting values in the study and those of (Pisano et al. 2002)

Parameter	Value	Pisano et al. (2002)
Recession velocity, v_{mid} (km s ⁻¹)	1975 ± 1.4	1986
Line width W , (km s ⁻¹)	170 ± 2.5	133
Flux integral, $\int S dv$ (Jy km s ⁻¹)	73.1 ± 5.0	13
Distance, D , (Mpc)	28.2 ± 0.4	31
Hydrogen mass, M_{HI} , ($10^{10} M_{\odot}$)	1.38 ± 0.1	0.3
Dynamical mass, M_{dyn} , ($10^{10} M_{\odot}$)	5.60 ± 0.1	>2

compared my resulting plot and Fig. 6 of (Pisano et al. 2002). For the purposes of this paper, I just used a multiplicative factor of 21 to better match up my result with (Pisano et al. 2002). To make the plot look better, I took the average of the plot to transform it into a continuous smooth line.

To find the midpoint of the emission line, I first wrote code that plotted a spotted horizontal black line at the standard deviation of the plot ($y = \sigma = 0.03$). To separate the HI velocity data from the noise, I plotted two spotted vertical red lines at where the plot crosses the black line and printed the corresponding velocity values. I used

$$v_{mid} = v_{max} - v_{min} \quad (2)$$

where v_{max} is the greater velocity, v_{min} is the lesser velocity, and v_{mid} is the velocity corresponding to the midpoint of the spectral line.

3 RESULTS

In this study, I resulted in these values. For the recession velocity (v_{mid}), I received a value of 1975 ± 1.4 km s⁻¹. For the line width (W), I got a value of 170 ± 2.5 km s⁻¹. For the flux integral ($\int S dv$), I got a value of 73.1 ± 5.0 Jy km s⁻¹.

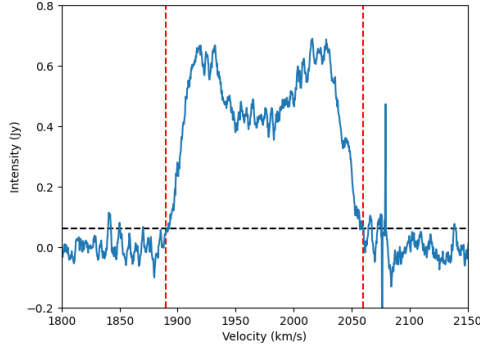


Figure 1. Figure 1: plot of the Intensity vs. Velocity. The black line is the standard deviation ($y = \sigma = 0.03$) and the vertical red lines are where the graph crosses y the standard deviation line.

4 DISCUSSION

Using the data collected in the results, I calculated more parameters revolving the Galaxy UGC 238. First, I used

$$D = \frac{v_{mid}}{H_0}, H_0 = 70 \frac{km/s}{Mpc}$$

where D is the distance to the Galaxy and H_0 is Hubble's constant. To find the dynamical mass of UGC 328, I used

$$M_{dyn} = 5.6 M_{\odot} \frac{W^2 R}{\sin(i)} \quad (3)$$

where $M_{\odot}$ is the mass of the sun, W is the spectral line width in $km\ s^{-1}$, R is the radius of the galaxy in pc ($R = 20.94$ pc), and i is the inclination angle between the plane of the galaxy and the plane of the sky. For this equation, I used

$$W = v_{max} - v_{min} \quad (4)$$

to find W , and

$$i = \arccos\left(\frac{1}{A_r}\right), A_r = 13/8 \quad (5)$$

where A_r is the axial ratio. I then compared the dynamical mass to the HI mass, which can be calculated by

$$M_{HI} = 2.36 \times 10^5 M_{\odot} \left(\frac{D}{Mpc}\right)^2 \int S dv \quad (6)$$

where S is the flux density in Jy and v is the velocity in $km\ s^{-1}$. To find the integral of the flux density, I multiplied the difference in on-spectrum and off-spectrum velocity values by the sum function of the plot. For the distance to the galaxy (D), I got a value of 28.2 ± 0.4 Mpc. For the inclination angle (i), I got a value of 0.91. For the hydrogen mass (M_{HI}), I got a value of 1.38 ± 0.1 ($10^{10} M_{\odot}$). For the dynamical mass (M_{dyn}), I got a value of 5.60 ± 0.1 ($10^{10} M_{\odot}$).

5 CONCLUSIONS

After comparing the results, I think my results compare well with the values recorded in the reference paper (Pisano et al. 2002). Even though the values are not exactly the same, they show resemblance between each other. I personally say this is a successful comparison study of the Galaxy UGC 328, however, future work is always needed for us to progress in the realm of astronomy. Some future work that I would personally like to see is HI studies into multiple Galaxies that are comparable to UGC 328. This study would compare all of

the received values from these Galaxies and try to infer what causes them to be different or similar to UGC 328.

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